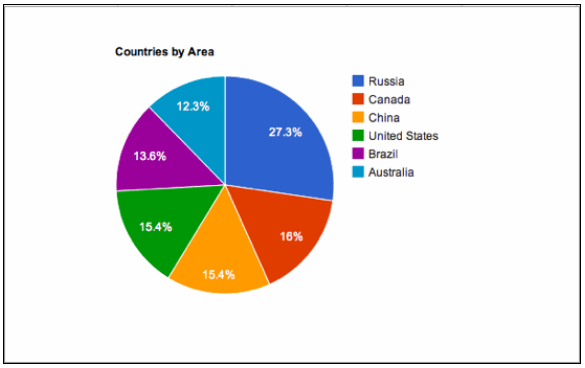
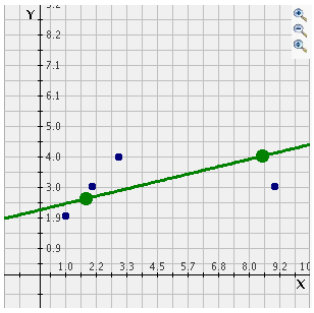
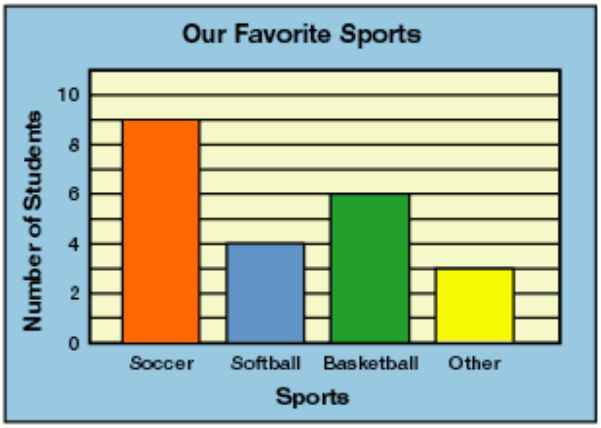
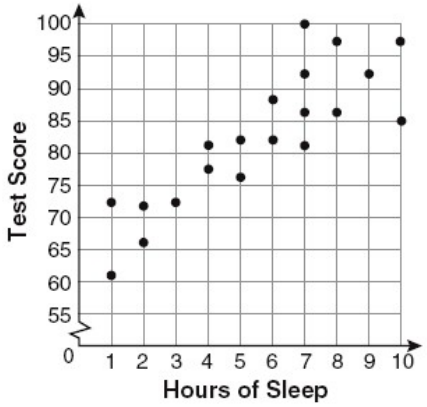


1-Variable Statistics & 2-Variable Statistics







One and Two Variable Statistics Test Outline

The test will be on _____. Graphing Calculators will NOT be permitted. Formulas will be provided but not labeled so you must be able to differentiate between them. (All formulas will be provided including those given on notes and in the textbook.)

Unit Overview

Topics for this test include, but are not limited to, the following:

1. Linear Correlation

- Response variable vs. explanatory variable
- Creating and interpreting scatter plots
- Outliers
- Covariance
- Coefficient of correlation
- Least squares methods
- The line of best fit/regression line
  - i. Finding the coefficients a and b
  - ii. Interpreting the values of a and b
  - iii. Graphing the line
- Residuals
  - i. calculating the residuals
  - ii. significance of residuals with regards to the regression line

2. Cause and Effect

- Various case and effect relationships

3. One-Variable Statistics

- Create a frequency table
- Create frequency, cumulative frequency, relative frequency diagrams
- IQR, box-and-whisker plots, outliers
- Discrete/continuous variables
- Working with grouped data/categorical data
- Measures of central tendency (calculate mean and weighted mean)
- Measures of spread
- Percentiles
- Z-scores

Textbook Questions	p. 212 #1-4, 7-13	p. 151 #1-3	p. 154 #1-6
	p. 214 #1-6	p. 152 #11-15	
	p. 219 #11, 12, 14-16	p. 153 #16-20	

Lesson: Organizing and Graphing Data

Raw Data

- the unprocessed information collected for a study

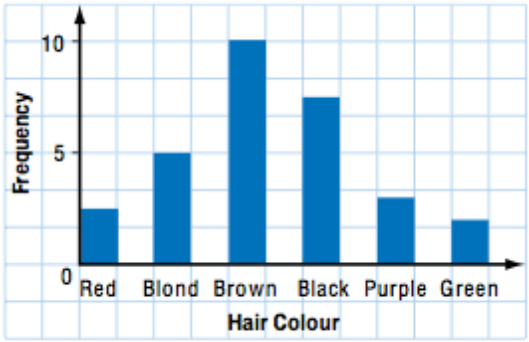
Example from Textbook Page 92:

Here are the sums of the two numbers from 50 rolls of a pair of standard dice.

11	4	4	10	8	7	6	6	5	10	7	9	8	8
4	7	9	11	12	10	3	7	6	9	5	8	6	8
2	6	7	5	11	2	5	5	6	6	5	2	10	9
6	5	5	5	3	9	8	2						

Bar Graph

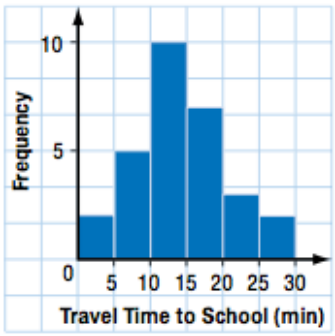
- a chart or diagram that represents quantities with horizontal or vertical bars whose lengths are proportional to the quantities
- represents all kinds of variables that may not have set order such as hair colour or citizenship



Bar Graph

Histogram

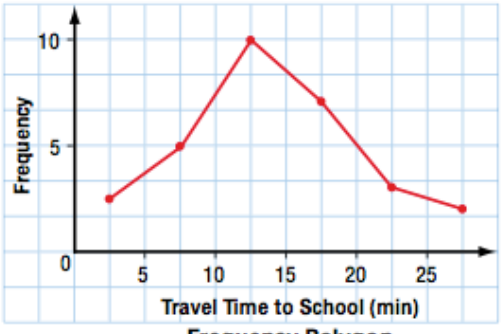
- a special form of bar graph which the areas of the bars are proportional to the frequencies of the values of the variable
- used for variables whose values can be arranged in numerical order especially continuous variables such as weight, temperature



Histogram

Frequency Polygon

- illustrates the same information as a histogram or bar graph by plotting frequencies versus variable values and then joining the points with straight lines



Frequency Polygon

Frequencies:

Example:

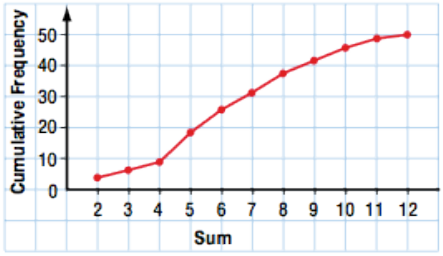
Here are the sums of the two numbers from 50 rolls of a pair of standard dice.

11	4	4	10	8	7	6	6	5	10	7	9	8	8
4	7	9	11	12	10	3	7	6	9	5	8	6	8
2	6	7	5	11	2	5	5	6	6	5	2	10	9
6	5	5	5	3	9	8	2						

Sum	Tally	Frequency	Cumulative Frequency
2	IIII	4	4
3	II	2	4+2=6
4	III	3	6+3=9
5	𝍸 IIII	9	9+9=18
6	𝍸 III	8	18+8=26
7	𝍸	5	26+5=31
8	𝍸 I	6	31+6=37
9	𝍸	5	37+5=42
10	IIII	4	42+4=46
11	III	3	46+3=49
12	I	1	49+1=50

Always the same as the number of rolls!!

Cumulative Frequency Graph will always be INCREASING (⇧)



Relative Frequency – shows the frequency of a data group as a fraction or percent of the whole data set

Example from Textbook Page 97: Marks on data management test:

78	81	55	60	65	86	44	90
77	71	62	39	80	72	70	64
88	73	61	70	75	96	51	73
59	68	65	81	78	67		

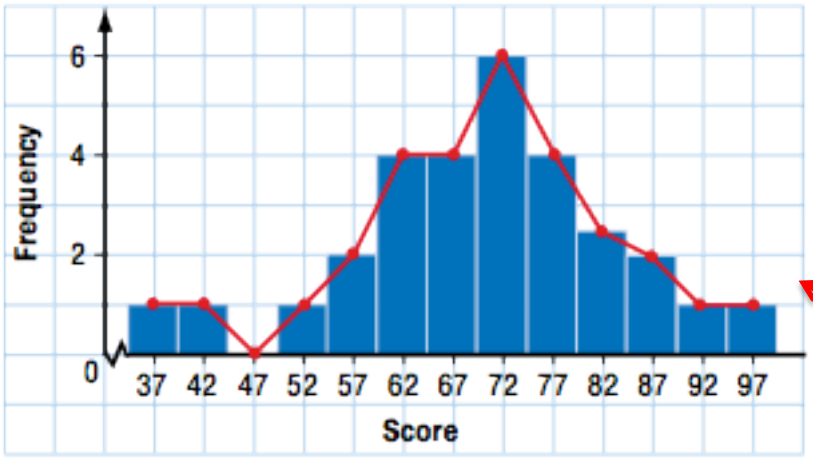
$$\textbf{interval size} = \frac{\textbf{possible range of data}}{\textbf{\# of intervals}}$$

$$\text{e.x. } \frac{96-39}{13} = 4.38 \backsim 5$$

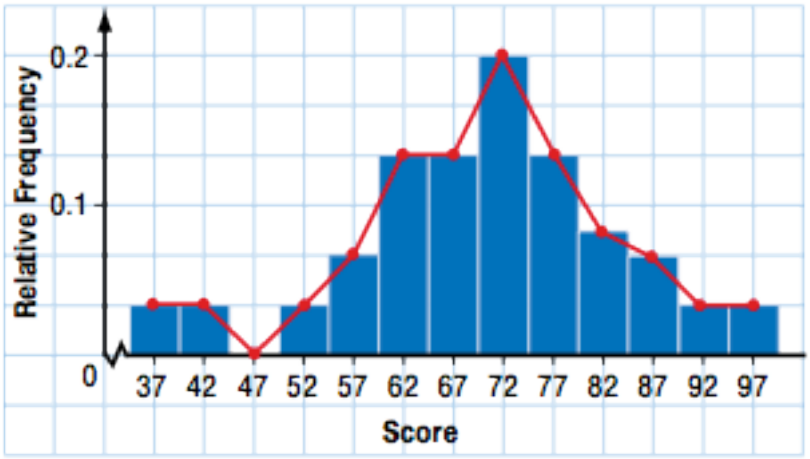
Marks(%)	Midpoint	Tally	Frequency	Relative Frequency (round to 3 decimal places)
34.5-39.5	37	I	1	$1 \div 30 = 0.033$
39.5-44.5	42	I	1	0.033
44.5-49.5	47	-	0	0
49.5-54.5	52	I	1	0.033
54.5-59.5	57	II	2	0.067
59.5-64.5	62	IIII	4	0.133
64.5-69.5	67	IIII	4	0.133
69.5-74.5	72	𝍸 I	6	0.200

74.5-79.5	77	IIII	4	0.133
79.5-84.5	82	III	3	0.100
84.5-89.5	87	II	2	0.067
89.5-94.5	92	I	1	0.033
94.5-99.5	97	I	1	0.033
Total			30	

Superimposed frequency polygon and histogram:



Superimposed relative-frequency polygon and histogram:



They should look exactly the same graph except the y-axis is different

Circle Graph

Example from Textbook Page 98:

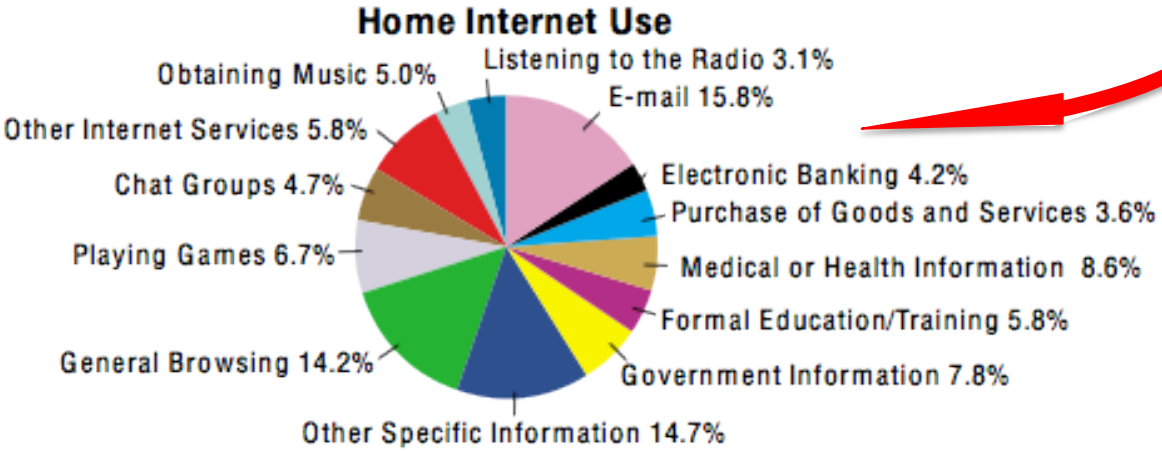
Primary Use	Households (%)
E-mail	15.8
Electronic banking	4.2
Purchase of goods and services	3.6
Medical or health information	8.6
Formal education/training	5.8
Government information	7.8
Other specific information	14.7
General browsing	14.2
Playing games	6.7
Chat groups	4.7
Other Internet services	5.8
Obtaining music	5.0
Listening to the radio	3.1

These percentages add up equal to 100% (one full circle)

One Circle = 360°

$1\% = 3.6°$

if 15.8%, then $15.8 \times 3.6° = 56.88°$ of your circle graph



Pictograph:

Home Internet Use

E-mail	💻💻💻💻💻💻💻💻
Electronic Banking	💻💻
Purchase of Goods and Services	💻💻
Medical or Health Information	💻💻💻💻
Formal Education/Training	💻💻💻
Government Information	💻💻💻💻
Other Specific Information	💻💻💻💻💻💻💻
General Browsing	💻💻💻💻💻💻💻
Playing Games	💻💻💻
Chat Groups	💻💻
Other Internet Services	💻💻💻
Obtaining Music	💻💻
Listening to the Radio	💻

Each 💻 represents 2% of households.

Organizing and Graphing Data Practice

Recall:

Continuous Data

$$interval\ size = \frac{possible\ range\ of\ data}{\#\ of\ intervals}$$

Relative Frequency Distribution

1. Given the final marks for a Grade 11 Math class.

42	44	45	50	51	53	55	56	57	59
60	61	62	62	64	64	64	66	66	67
68	69	71	71	72	74	75	81	84	89

a) Construct a frequency table that includes the column for the relative frequency

Marks (%)	MIDPOINT	TALLY	FREQUENCY	RELATIVE FREQUENCY

b) Construct a histogram and frequency polygon



c) What proportion of students had marks between 60% and 69% exclusively?

Lesson: Measures of Central Tendency

When you analyze data that has been collected, the first measure to find is the "average" result for the collection.

There are three measures of central tendency that we can calculate.

- Mean – add all the data and divide by the number of data items.
- Median – the middle piece of data when the items are ranked from least to greatest
- Mode – the most frequencyly occurring data value

The symbol used for sample mean is $\bar{x}$.

The symbol used for population mean is μ.

Calculations:

The formula used to calculate the sample mean is $\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$, where n represents the total number of values and x represents the observed values.

Example: Find the mean of the following data. 4,9,4,5,3,8,10,12

To calculate the median, arrange the data values for least to greatest. Calculate $\frac{n+1}{2}$, where n represents the number of values, to determine the location of the middle value. If the data presents an even number of values, average the two values in the middle position.

Example: Find the median of the following data. 4,9,4,5,3,8,10,12

To find the mode, determine the most frequent value. It is possible for there to exist more than one mode.

Example: Find the mode of the following data. 4,9,4,5,3,8,10,12

A weighted mean is calculated when some of the data items in an average are worth more than others.

The formula used to calculate a weighted mean is $\overline{x_w} = \frac{\sum w_i x_i}{\sum w_i}$.

An example of a weighted mean is the mark breakdown of our courses. Your final mark is MDM4U1 will be based on a weighted mean which is based on categories.

Example:

The overall performance of a student in Data Management was recorded as follows:

CATEGORY	WEIGHT	STUDENT MARK %
Knowledge and Understanding	24%	75%
Application	18%	70%
Thinking	14%	60%
Communication	14%	80%
Culminating Task	15%	70%
Final Exam	15%	55%

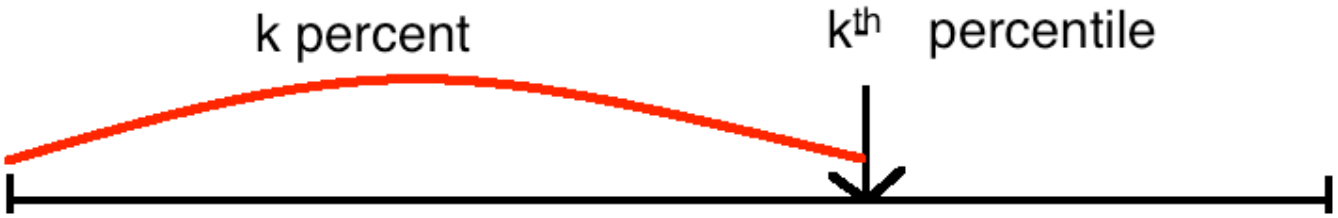
a) Calculate the student's term mark. The term mark does not include the culminating task or the final exam.

b) Calculate the student's Summative Evaluation mark. This mark only includes the culminating task and the final exam.

c) Calculate the student's final mark.

Learning Goal: Measures of Spread

PERCENTILE – divide the data into 100 intervals that have equal numbers of values. It is a measure that indicates what percent of the given population scored at or below the measure.



To calculate a particular percentile:

1) Multiply the total number of values in the data set by the percentile which will give you the index.
2) Order all of the values in the data set in ascending order (least to greatest)
3) If the index is a whole number, count the values in the data set from least to greatest until you reach the index, then take the index and the next greatest number and find the average.
4) If the index is not a whole number, round the number up, then count the values in the data set from least to greatest until you reach the index.

Textbook page 145:

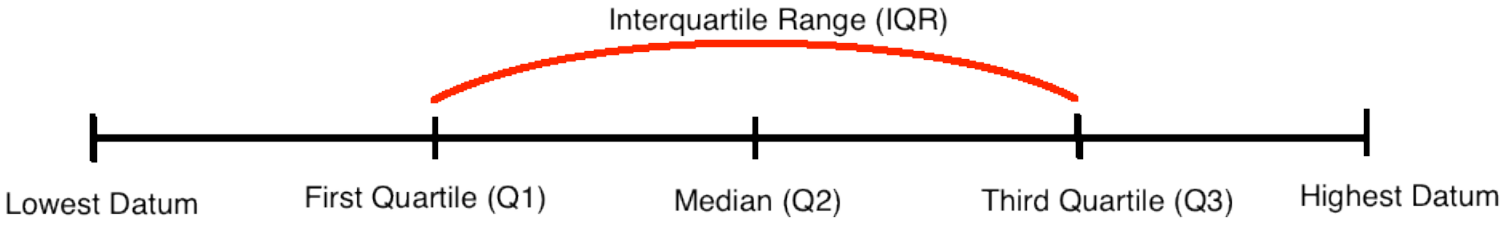
35	47	57	62	64	67	72	76	83	90
38	50	58	62	65	68	72	78	84	91
41	51	58	62	65	68	73	79	86	92
44	53	59	63	66	69	74	81	86	94
45	53	60	63	67	69	75	82	87	96
45	56	62	64	67	70	75	82	88	98

a) If a datum scored at 50th percentile, what was its raw score?

b) What is the 90th percentile for this data?

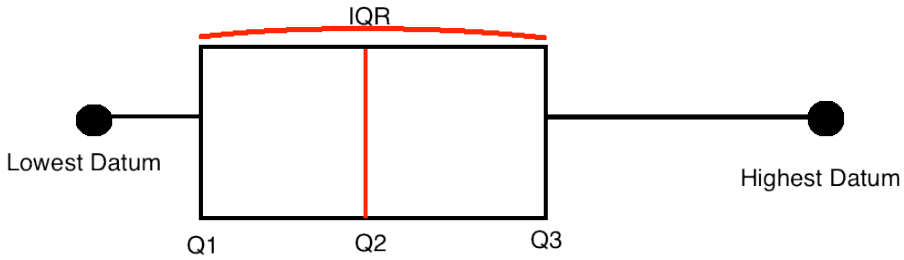
c) Does the score of 75 place it at the 75th percentile?

Quartile and Interquartile Range (IQR):

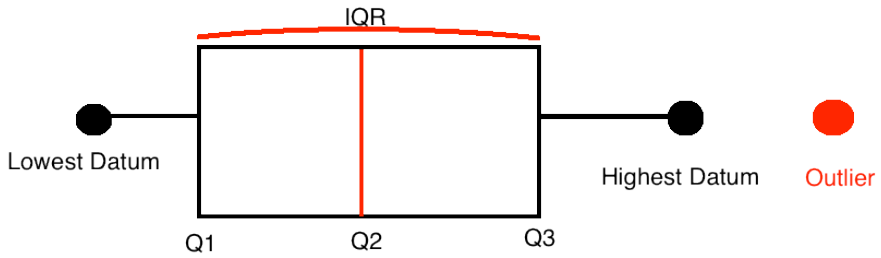


$$Q2 = Median$$
$$Q1 = Median\ of\ the\ first\ half\ of\ the\ data$$
$$Q3 = Median\ of\ the\ second\ half\ of\ the\ data$$
$$IQR = Q3 - Q1$$
$$SemiIQR = SIQR = \frac{IQR}{2}$$

Box-And-Whisker Plot:



Modified Box-And-Whisker Plot:



To determine the value(s) of an outlier:

$$1.5 \times IQR$$

If a datum is less than $Q1 - (1.5 \times IQR)$, it is considered as an outlier.
If a datum is greater than $Q3 + (1.5 \times IQR)$, it is considered as an outlier.

Example #1: A survey of movie goers at a screening of "Rocky Horror Picture Show" asked how many times they have seen the movie. The results for 20 respondents were:

3 4 2 8 10 5 1 15 5 16 6 3 4 9 12 3 30 2 10 7

a) Find the median, Q1, and Q3.

b) Calculate IQR and SIQR.

c) Are there any outliers? Explain.

d) Make a box-and-whisker plot of the results.

e) Make a modified box-and-whisker plot of the results accounting for any outliers. How do the quartiles compare?

Using the previous example, find the score of:

a) 20th percentile
b) 65th percentile
c) 95th percentile

Learning Goal: Measures of Spread

Deviation is the difference between an individual value in a set of data and the mean for the data.

x = the individual value in a set
μ = the population mean
$\bar{x}$ = the sample mean

For the population, deviation is $x - \mu$

For a sample, deviation is $x - \bar{x}$

The larger the size of the deviations, the greater the spread in the data.

Standard Deviation – measure of variability (How much variation exists from the average)

- low standard deviation indicates the data points are very close to the mean
- High standard deviation indicates the data points are very far from the mean

Variance - how spread out numbers are

Measure	Sample	Population
Standard Deviation	$s = \sqrt{\frac{\sum(x-\bar{x})^2}{n-1}}$ n is the number of data in sample	$\sigma = \sqrt{\frac{\sum(x-\mu)^2}{N}}$ N is the number of data in population
Variance	$s^2 = \frac{\sum(x-\bar{x})^2}{n-1}$	$\sigma^2 = \frac{\sum(x-\mu)^2}{N}$

Z-Scores - the number of standard deviation that a datum is from the mean

Sample	Population
$z = \frac{x-\bar{x}}{s}$ s=standard deviation of the sample	$z = \frac{x-\mu}{\sigma}$ σ = standard deviation of the population

- Variables below mean have NEGATIVE Z-SCORES
- Variables above mean have POSITIVE Z-SCORES
- Variables equal to mean have ZERO Z-SCORE

Ex. #1 -the number of hours five students spent studying statistics last week: 8 4 9 11 3

a) Calculate Mean

b) Calculate Variance

c) Calculate Standard Deviation

Ex. #2 – The number of summer jobs a sample of six students applied for: 17 15 23 7 9 13
Calculate the Standard Deviation

Ex. #3 – Comparing Consistency of Two Types of Golf Clubs Consistency is the hallmark or a good golfer. Golf equipment manufacturers are constantly seeking ways to improve their products. Suppose that a recent golf innovation is designed to improve the consistency of its users. As a test a golfer was asked to hit 150 shots using a 7-iron, 75 of which were hit with his current club and 75 with the new innovation 7-iron. The distances were measured and recorded. Which 7-iron is more consistent?

Current	
Mean	150.55
Standard Deviation	5.79
Sample Variance	33.55

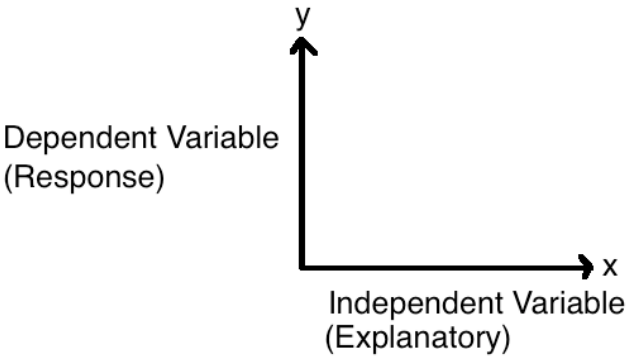
Innovative	
Mean	150.15
Standard Deviation	3.09
Sample Variance	9.56

Ex. #4 – If the mean is 68.1 and the standard deviation is 15.2, determine the z-scores for Audio Maximizer Ultra 3000 (score 67) and SchmederVox (score 75).

Lesson: Linear Correlation and Regression

Scatter Plot

- graph that describes relationship between two variables

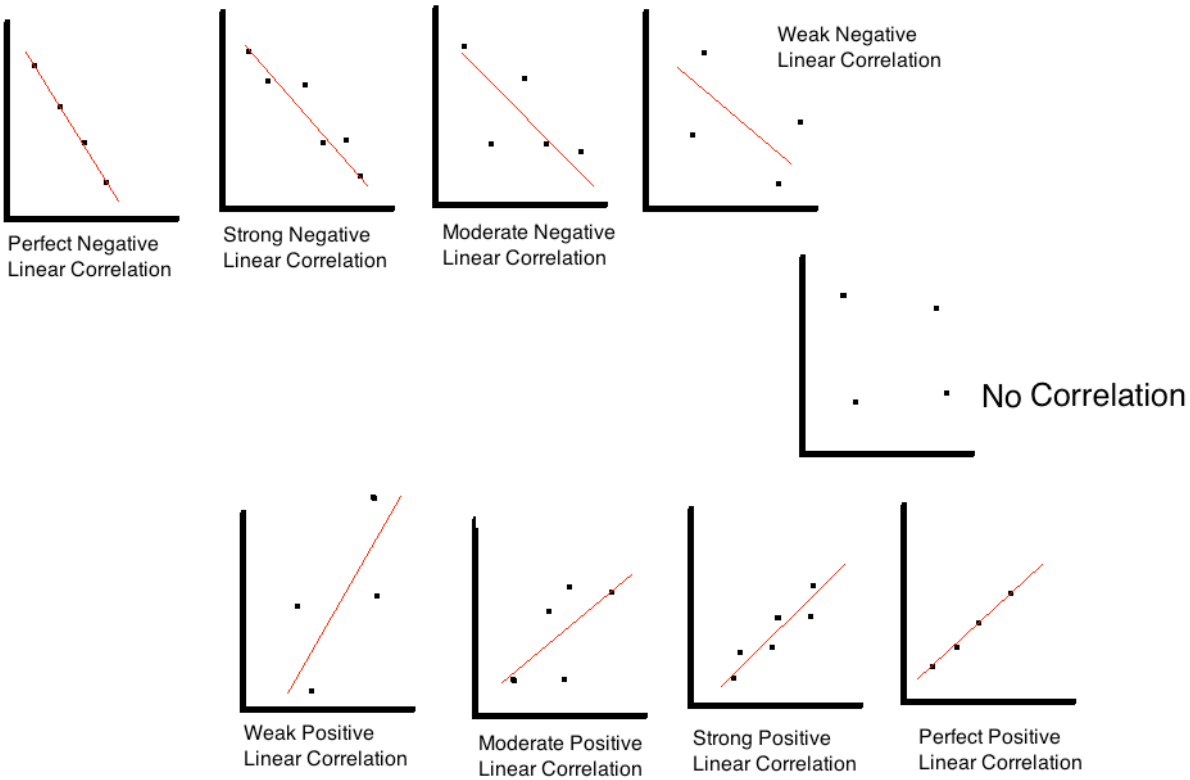


Line of Best Fit

- passes as close as possible to all the points on a scatter plot and represents the relationship between two variables

Linear Correlation

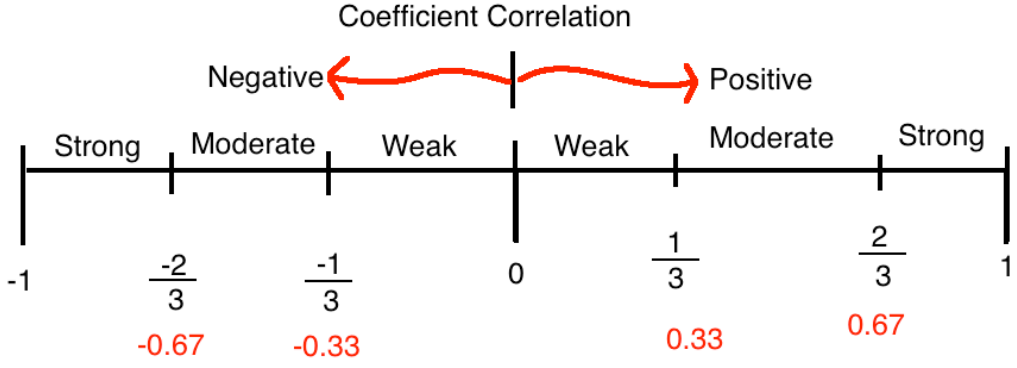
- changes in one variable that are proportional to changes in another
- weak, moderate, or strong (strength of the relationship)
- positive or negative (direction of the relationship)



Coefficient of Correlation

- the strength between the two variables

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2]\,[n\sum y^2 - (\sum y)^2]}}$$



Linear Regression

- a technique to find $y = ax + b$ for the line of best fit
- "Least Square Method"

$$a = \frac{n\sum xy - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

$$b = \bar{y} - a\bar{x}$$

Covariance

- measure of correlation which reveals whether the relationship between the variables is positive or negative

$$S_{xy} = \frac{1}{n-1}\left(\sum xy - \frac{\sum x \sum y}{n}\right)$$

Residual

- the difference between the values of y at the data point and at the point that lies on the line of best fit and has the same x-coordinate as the data point

$$Residual = observed - predict$$

positive = data above the line of best fit
negative = data below the line of best fit

Coefficient of Determination for non-linear regression

- r^2 (the square of the correlation coefficient)
- it measures the proportion of the variation in y that is explained by the variation in x
- it tells what percentage of points that are on the line/curve of best fit
- $0 \leq r^2 \leq 1, r^2 = 1$ the curve is a perfect fit
- this applies to any type of regression

Linear Correlation and Regression Practice **Date:**

Determine the linear regression and coefficient of correlation for the following 2 examples.

Example #1

x	y	x^2	y^2	xy
2	13			
6	20			
7	27			
$\sum x$	$\sum y$	$\sum x^2$	$\sum y^2$	$\sum xy$

Example #3:

Are the marks one receives in a course related to the amount of time spent studying the subject? To analyze this possibility, a student took a random sample of 10 students who had enrolled in an accounting class last semester. She asked each to report his or her mark in the course and the total number of hours spent studying accounting. These data are listed here.

Hours spent	40	42	37	48	25	44	41	48	35	28
Marks	77	63	79	86	51	78	83	90	65	47

a) Calculate the covariance.

$$S_{xy} = \frac{1}{n-1}\left(\sum xy - \frac{\sum x \sum y}{n}\right)$$

b) Calculate the coefficient of correlation.

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

c) What do the statics you have calculated tell you about the relationship between marks and study times?

Example #4:
Least Squares Method for Finding Equation of a Line of Best Fit

Age (years) - (X)	Annual Income ($) - (Y)	x^2	y^2	xy
33	33			
25	31			
19	18			
44	52			
50	56			
54	60			
38	44			
29	35			
$\sum x$	$\sum y$	$\sum x^2$	$\sum y^2$	$\sum xy$

1) Substitute these totals into the formula for a.

$$a = \frac{n\sum xy - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

2) To determine b, you also need the means of x and y.

3) Now substitute the values of a and b into the equation:

4) You can use the equation of the line of best fit as a model.

5) Predict the income for an employee who is 21 and an employee retiring at age 65.

a) For a 21 year old employee,

b) For a 65 year old employee,

6) Determine which one is interpolation and which one is extrapolation from question #5.

7) Comment on the accuracy of both estimates.

Lesson: Cause-and-Effect Relationship

5 different types and degrees of causal relationship between variables:

1) Cause-and-Effect Relationship
   A change in x produces a change in y

2) Common-Cause Factors
   External variables causes two variables to change in the same way

3) Reverse Cause-and-Effect Relationship
   The dependent and independent variables are reversed in the process of establishing causality

4) Accidental Relationship
   A correlation exists without any causal relationship between variables that happens by random chance

5) Presumed Relationship
   A correlation does not seem to be accidental even though no cause-and-effect relationship or co mmon cause factor is apparent.

Try This… Classify the relationships in the following situations.

a) The rate of a chemical reaction increases with temperature

b) Leadership ability has a positive correlation with academic achievement

c) The price of butter and motorcycles have a strong positive correlation over many years

d) Sales of cellular telephones had a strong negative correlation with ozone levels in the atmosphere over the last decade

e) Traffic congestion has a strong correlation with the number of urban expressways

More……..

Extraneous Variables

- Variables that affect or obscure the relationship between an independent and a dependent variable

Experimental Group

- The group for which the independent variable is changed in an experiment or statistical study

Control Group

- The group for which the independent variable is held constant in an experiment or statistical study

Example 1:A medical researcher wants to test a new drug believed to help smokers overcome the addictive effe cts of nicotine.
• One hundred people who want to quit smoking volunteer for the study.
• The researcher carefully divides the volunteers into two
groups, each with an equal number of moderate and heavy smokers.
•One group is given the nicotine patches with the new drug, while the second group uses ordinary nicotine pat ches.
• Twenty-eight people in the first group quit smoking completely, as do eighteen people in the second group.

a) Identify the experimental and control groups.

EVALUATING STATISTICAL CLAIMS:

Factors leading to Erroneous Conclusions

- Bias in the survey
- Outliers in the data
- Failing to account for extraneous variables
- Failing to detect hidden variables
- Assuming a strong correlation proves the existence of a cause-and-effect relationship

Assessing methods of collecting and analyzing data

- Is the sample process free from bias?
- Could outliers or extraneous variables influence the results?
- Are there any unusual patterns that suggest presence of hidden variables?
- Has causality been inferred with only correlation evidence?